

10. Multi-Threshold Segmentation

Pixels: [20, 60, 120, 180]

(a) Interpretation of multi-level segmentation

Multi-level (multi-threshold) segmentation classifies pixels into more than two classes by applying several threshold values, rather than a single global threshold that only produces a binary (object/background) result. This is useful when an image contains more than two meaningfully different intensity regions.

(b) Justification for thresholds (50, 100)

These thresholds are chosen where the image histogram shows valleys — natural dividing points between clusters of pixel intensities — so that 50 separates the darkest class from the mid-tone class, and 100 separates the mid-tone class from the brightest class.

(c) Apply segmentation

Rule: Class 1 (dark) if $\text{pixel} < 50$; Class 2 (mid) if $50 \leq \text{pixel} < 100$; Class 3 (bright) if $\text{pixel} \geq 100$.

(d) Computed classes

Pixel value	20	60	120	180
Assigned class	Class 1 (dark)	Class 2 (mid)	Class 3 (bright)	Class 3 (bright)

(e) Interpretation

The four pixels are separated into three meaningful intensity classes — 20 is grouped as dark background, 60 as a mid-tone region, and 120 and 180 together as the bright/foreground region — giving a richer segmentation than a simple two-class (binary) threshold would.